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Research assignment 1

(1) A database is an organized collection of information that is stored electronically. This makes it easy for the information to be queried-(searched, updated, managed and shared)

Data bases are essential in the modern day world because if the information stored is processed/queried diligently this can offer deeper insights to whatever the data base may be related to. Businesses can make informed decisions, School can plan better and so forth.

Data bases are a major part of our daily lives. They help different entities /organizations - store, organise and use information efficiently, these can be ~~government~~ goverment organizations, businesses and even a mom shopping for her family as long the information is structured it can unveil alot.

(2) A Data Warehouse is a large storage system that collects and stores data from different Sources in one central place.

- Stores large amount of historical data.
- Used for analysis or reporting.
- Updated in batches at a time.
- Focuses on analysis of large data sets.

A company will use a data warehouse to:
- Analyze business performance.
- Combine data from different departments
- Identify trends and patterns.
- Improve decision making.

(3) A data warehouse stores organized data for reporting and decision making, while a data lake stores large amounts of raw data for flexible analysis and future use. Businesses choose data lakes when they need to manage huge and different types of information.

(4) A data-mart is like a slice of a pie within a data warehouse - the data warehouse being the entire pie. A smaller section that is designed for a specific business function or department. eg Sales, finance or marketing - This helps these specific departments access relevant data quickly and efficiently.

(5) Structured-Data is organized and stored in a fixed format eg: rows and columns in a table. This makes it easy to manage. A good example is an electronic register for attendance with all the attendees information

Semi-Structured-data is partly organized but would not fit into a table eg: an email.

Unstructured-Data has no fixed format or organisation method. Difficult to store and analyze eg: Social media (~~photho~~ photos)

⑥ Data modeling is the process of creating a visual and logical design of how data will be stored, organized, connected and used in a system. This in simple terms is similar to drawing a house plan before building.

Data modeling
- Improves organisation of data.
- Reduces errors and duplication.
- Maked databases easier to understand.
- Improves system performance.
- Helps developers and businesses communicate clearly
- Ensures accurate reporting & analysis.

One example is a – Logical Data Model
→ This type of data model provides detail about ~~the~~ the structure of the data, including attributes and relationships but is still ~~to~~ independent of the actual database software

Example

A customer info table on a excel document.

Customer-ID	Name	Email adress	Contact Number
0034	Sophie Ndaba	Sophie@yahoo.com	081 442 1498
0035	Richard Chauke	R.Chauke@.....	011 267 2020
0036	Omphile lekoko	O.email@work	021 448 6741
0037	Rose Nkosi	Nkosi R@gmail.com	072 419 0589

Another example is a Conceptual Data model
- a conceptual data model gives a high-level ~~over~~ over-view of the data system. It focuses on the main enitities and relationships without technical details

Example
- in a School system
① * Classed attended by students.
② * Subjects taught by teachers.

7) Data mining is an important process that helps businesses discover insightful information hidden inside large data sets. Companies use it to improve services, increase profits, understand customers and reduce risk.

→ Companies like take-alot and Amazon use data mining to analyze customer purchase and browsing history, identifying shopping patterns and suggest products customers are likely to buy to improve sales and customer experience.

→ Fraud Detection in banking
Large banks use data mining to detect suspicious transactions. Bank analyze spending patterns to protect customer accounts, improving security and reducing finacial loses.

8) A Data Management system is a software used to collect, store, organize, manage and retrieve data efficiently.
This helps organisations control their data securely and ensure information is accurate and easy to access.

These systems are important because they help organizations
* Store large amounts of data.
* Analyze and manage business operators.
* Prevent data loss
* Improve security
* Share information easily

1) My SQL
- a popular relational database management system that stores data in tables using rows and columns using structured query language to manage data.

→ Fast & reliable
→ Easy to use
Perfect for a small online shopping website where customer details, product information, orders, payments ect....

② MongoDB
- is a NoSQL database that stores data in a flexible document instead of traditional tables.
- Handles large amounts of unstructured ~~unstored~~ data.
- Flexible and scalable.
- Good for modern web and mobile apps for example social media platforms can use MongoDB to store:
- Use profiles
- Posts
- Comments
- Photos and videos

③ Oracle
Oracle is an enteprise-level database system and used by large businesses / organizations worldwide.
- High security
- Very reliable
- Handles massive data volumes.
- Strong back ~~up~~ up and recovery.

Suitable for large hospitals and government departments to manage
- Patient records
- National data bases
- Large business systems

Microsoft SQL Server
- is a powerful database platform developed by microsoft for storing and managing business data
- strong security.
- advanced reporting tools.
- Good performance for large businesses.

Banks can use SQL server to manage:
- Finacial records
- Employee information.
- Business reports.

(9) This process used in data management and data analytics to collect data from different sources, clean and organise the data and then store it in a data-base or data warehouse for analysis. ETL helps businesses turn raw data into useful information for reporting ~~or~~ and decision-making.

(1) Extract

This step involves collecting data from data sources for example
- Databases
- Excel files
- Sales systems ect...

* Modern business normally extract from cloud services such as google-cloud ect
* Manual extraction.

The importance of this step is to gather all needed data in one place.

(2) Transform

This step clears and prepares the data so it becomes accurate and useful.
- correcting errors
- Removing duplicate
- Changing data formats
- Combining data from different systems

This step helps improve data quality.

(3) Load

This step stores the cleaned and transformed data into a final system such as

- A data-base.
- A data warehouse.
- A data lake.

ETL is a Key process in data analytics that helps organizations clean, organize data for analysis. The above mentioned steps ensure organizations work with reliable information to make smarter decisions.

10) 1) CSV (Comma-seperated values

- A CSV file stores data in plain text using commas to seperate values.
- Commonly used in microsoft excel, Data imports & exports.

2) JSON (JavaScript Object Notiation)

- Stores data in Key-value pairs and is commonly used for web and API data.
- Commonly used in web applications, cloud systems.

3) XML (Extensible Markup language)

- Stores data using tags to describe information
- Commonly used in banking systems and Government systems.

4) Parquet

- is commonly used in data analytics.
- Commonly used in Data lakes, cloud analytics systems

5) Avro

- is a compact data format used in big data systems

Diffent storage formats are suited to different analytics needs. CSV and JSON are common for smaller systems and web applications. While AVRO and Parquet are applications best suitable for bigger data and advanced analytics enviroments. Choosing the correct format improves performance, storage, effiency and data processing speed.

(11) AI is the ability of computers or machines to perform tasks that normally require human intelligence such as:
- Learning
- Problem solving
- Understanding language
- Decision making ect....

* These systems use data and algorithms to imitate human-thinking and behaviour.

NARROW AI (Weak AI)	General AI (STRONG AI)
Designed for one specific task	Designed to perform many diffent tasks like humans
Cannot think beyond its programmed function.	Can learn, reason, adapt across many situations
Exists today and widely used	Still mostly theoretical and not fully developed
Limited intelligence	Human like intelligence
Example google maps.	There is currently no fully developed real-world general AI

(13) Deep learning is a special branch of machine learning that uses artificial neural networks to learn from very large amounts of data.
It is designed to solve complex problems such as
- Image recognition
- Speech "
- Language translation
- Self-driving cars

Deep learning models can learn automatically without needing as much manual programming

Neural Networks

- are computer systems inspired by the human brain
- They consist of connected layers of artificial neurons that process information and learn patterns from data.
- A neural network usually has:
  1. * Input Layer — receives information
  2. * Hidden Layers — processes and analysis data.
  3. * Output Layer — produces the final result

AI is the broad concept of machines performing intelligent tasks. Machine learning is a subset of AI where systems learn from data.
Deep learning is a subset of machine learning that uses deep neural networks.

(14) Natural Language Processing
- is a branch of AI that allows/enables computers to understand, interpret, process and respond to human language. This allows machines to work with both spoken and written languge in a way that feels natural to humans.
- It combines
  - Computer science
  - Artificial intelligence
  - Linguistics

to help computers understand language, patterns, meaning and communication

NPL systems can:
Understand speech, translate languages, recognise text, answer questions, predict words while typing and also analyse emotions in text.

Real-World applications
- Voice assistance - Eg Siri or Google Assist
- Translation Apps - Eg Google translate
- Chatbots on websites.
- Auto correct on smart phones

(15) A Large Language Model (LLM)
- is a type of AI that is trained on ~~a~~ huge amounts of text data so it can understand, generate and respond to human language

- LLMs learn patterns by analysing books, websites, articles and conversations.
- They can answer questions, write text, articles, translate languages, summarize information, generate code and hold conversations
- The models are called large because they are trained using massive datasets and billions of parameters

Standard Languge Model
- Is normally trained on smaller data sets
- Handles limited language tasks
- Less accurate and flexible
- Usually designed for one purpose
- Requires more human guidance

Some well know LLMs are ChatGPT and Gemini

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(16) Generative AI

- is a type of artificial intelligence that can create new content such as text, images, music, videos and code. In simple terms generative AI produce original content based on patterns it learned from large datasets.

- How it works is that it trains on massive amounts of data using machine learning and deep learning techniques.
  1. Trains on large data.
  2. Learns patterns using neural networks & deep learning
  3. When prompted the AI will predict and generate most suitable content

Some example of tools that use GI are:

1. ChatGPT
   - answers questions
   - writes text and essays
   - Helps coding ect...

2. DALL·E
   - generate images from text descriptions

3. GitHub Copilot
   - Assist programmers by generating code suggestions

(17) A training dataset is a collection of data used to teach an AI model how to recognize patterns and make decisions.
The quality of traing data is very important because AI learns directly from the data it receives. If the data is accurate and balanced, the AI will produce better and more reliable results

If the training data is biased or incorrect
- The AI may make unfair decisions.
- Predictions may be inaccurate.
- The system may produce errors.
- Certain groups of people may be treated unfairly.

(18) Computer Vision is a type of artificial intelligence that allows computers to see, analyse and understand images and videos.
- Machines interpret images by:
  ① Capturing images with cameras
  ② Converting images into digital data.
  ③ Using AI models and neural networks to recognize patterns, shapes and objects
  ④ Making decisions based on what they detect.

Some examples of computer vision use in the real-world are:
- Smart phones using facial recognition as a security measure.
- Telsa use computer vision in self driving cars to detect roads, traffic signs ect...

(19) An API (Application programming Interface
— Is a tool that allows different software systems or applications to communicate and share information with each other.

— An API is like a waiter in a restaurant:
① You place an order.
② Your waiter take your order to the Kitchen.
③ The Kitchen prepares the food.
④ The waiter brings the food back to you.

In simple terms an API connects two software systems like the waiter acts as the connection between you and the kitchen. A good example is how google maps API allows my bolt/Uber app to show maps and directions

In conclusion APIs help applications share data and services efficiently, making modern apps, website and digital systems work together smoothly.

(20) An API Key is a unique code used to allow authourized access to an API. Its purpose is to verify users or applications, protect data and services and track API usage.

API Keys should be Kept private, stored securely and never shared publicly. Developers often use encrypted storage, access controls and secret mangement tools.

If exposed publicly:
→ Hackers may misuse the API
→ Data could be stolen
→ Services may be abused
→ Finacial or Security damage could occur.

(21) A REST API is a type of API that allows different systems to communicate over the internet using standard HTTP methods. These are commonly used websites, mobile apps, cloud services ect--
- REST stands for Representational State Transfer
- A set of rules developers follow when designing web services and API's.

METHOD	PURPOSE	DESCRIPTION
GET	Retrieve data	Used to request/read data from a server
POST	Create data	Used to send new data server
PUT	Update data	Used to modify/replace data
DELETE	Remove data	Used to delete data from the server

(22) A JavaScript Object Notation
- is a lightweight data format used to store and exchange information between systems, applications and APIs.

JSON organises data using:
Key value pairs
→ Brackets { }
→ Arrays [ ]

JSON is popular because it is:
→ Easy for humans to read.
→ Easy for computers to process
→ Lightweight and fast.
→ Supported by many programming languages.
→ Ideal for sending data over the internet.

Example
```
{
"name": "John"
"age": 25,
"city": Johannesburg
}
```

→ "name" is the key and "John" is the value
→ "age" is the key and 25 is the value
→ "city" is the key and "Johannesburg" is the value.

23) Prompt engineering
- is a skill of writing clear and effective instructions (prompts) to help AI tools generate accurate and useful responses

It involves choosing the right words, details and structure when communicating with AI systems. Good prompts help AI to understand requests clearly, produce more accurate answers, save time, reduce confusion and errors, generate better-quality content.

Examples

"Write about dogs"	"Write a 100-word paragraph about why dogs make good pets for families"
"Help with coding".	"Explain how to create a python program that calculates the average of five numbers"

24 A MAP function

- is a programming function used to apply the same operation or change to every item in a list or collection of data.
- It "maps" each item to a new value and returns a new list with updated results.

MAP functions are commonly used in:

- Data processing.
- Data analytics
- Functional programming

because they make ~~good~~ code shorter and more efficient.

Python Example

```
numbers = [1, 2, 3, 4]
result = list (map (lambda x : x * 2,
print (result)
```

Output:

Plain text

[2, 4, 6, 8]

→ map () applies a function to every item in the list

→ lambda x: x * 2 (means multiply each number by 2)

→ The result is a new transformed list

25 Cloud computing

- is the delivery of computing services such as storage, serves, databases and software over the internet instead of using local computers or physical servers.
- It allows users access to technology services online from anywhere.

Three main Cloud service models

Model	Meaning	Description	Example
IaaS	Infrastructure as a service	Provides virtual serves, storage, and networking online. Users manage applications and operating systems themselves	Amazon web service
PaaS	Platform as a service	Provides a platform for developers to build, ~~trust~~ test, and deploy applications without managing hardware	Google App Engine
SaaS	Software as a Service.	Provides ready-to-use software over the internet. Users simply access the application online	Microsoft 365

26 A Version Control System (VCS)

- is a tool that tracks and manages changes made to files, code, or projects over time.
- It helps teams collaborate, restore previous versions and manage updates safely.

GIT

- is a popular distributed version control system used by developers and data analysts to manage code and project files.

GIT is widely used

- Track file changes
- Support teamwork and collaboration.
- Prevents overwriting working.
- Allows rollback to older versions
- Helps manage different project versions.

A Commit

- is a saved snapshot of changes made to a project on GIT
* Records what ~~~~ changed.
* Records when it changed.
* Who made the changes.

A Branch

- is a seperate version of a project where developers can work on new features or fixes without affecting the main project. Once changes are tested, the branch can be merged back into the main version.

(27) A Jupyter Notebook

- is an interactive tool used for writing and running code, analysing data and displaying charts or notes in one place

- It is commonly used with python. Jupyter notebook is popular because it is easy to use, Allows code, text and graphs together. It is also useful for data analysis and machine learning and also helps with testing and visualising data.

(28) Python

- is a high-level programming language known for simple, readable and powerful.

- It is popular in data analytics and AI because it is beginner friendly, has many data and AI libraries, handles large amounts of data efficiently and supports automation and machine learning.

Four Common Python libraries

1. Pandas - Used for data analysis & working with tables & data frames
2. NumPy - Used for mathematical caculations and arrays.
3. Matplotlib - Used for creating charts and graphs
4. Scikit - learn - Used for machine learning and predictive models.

29 Data privacy

- is the protection of personal and sensitive information from unauthorised access, misuse or sharing. It involves controlling how data is collected, stored, used and shared.

- Personal data includes information such as names, phone numbers, ID numbers, banking details ect ....

- Data privacy ~~is~~ is important becaus:

① Protects personal information.
② Prevents identity theft and fraud.
③ Builds trust between companies and users.
④ Helps with regulations and laws to be followed by organisations.
⑤ Protects confidential business information.

GDPR

- is a data privacy law created by European Union to protect peoples personal data and privacy
- It gives individuals more control over how companies collect and use their information
- GDPR applies to all countries in the European Union (EU). European Economic Area (EEA) countries.
- It also applies to companies outside Europe if the collect or process data belonging to EU citizens

(30) Data Ethics

- refers to the moral ~~principels~~ principles and rules that guide how data and AI should be collected, used, stored and shared responsibly and fairly

- It focuses on:
→ Privacy, fairness, transparency, accountability and responsible technology.

Examples ~~of~~ of Unethical Use of Data or AI

① Fake videos or deepfakes
- AI generated fake videos can spread information, damage reputations and manipulate people into believing false information

② Biased AI hiring systems.
- If an AI system is trained ~~to~~ on biased data it may unfairly reject certain groups of people, leading to discrimination in hiring.

③ Using personal data without consent.
- collecting or sharing peoples private information without ~~persi~~ permission violets privacy rights and can lead to identity theft or misuse data.

(31) Data governance

- is the process of managing data properly within an organisation to ensure it is accurate, Secure, consistent and used responsibly.

- It involves creating rules, policies and procedures for handling data throughout its life cycle. The goal is to make sure data is reliable, secure, well-organised and compliant with laws and regulations

→ Components of a good Data Governance Frame work

① Data quality
→ Ensures data is accurate, complete and consistent.

② Data Security
→ Protects data from unauthorised access, theft, or loss.

③ Policies and Standards
→ ~~Roles~~ Define rules for collecting, storing and using data.

④ Roles and Responsibilities
→ Assigns people or teams to manage and and maintain data

⑤ Compliance and Privacy
→ ~~End~~ Ensures data handling follows laws such as GDPR

⑥ Data management processes
→ Control how data is stored, updated, shared and deleted.

(32) Bias in AI

- happens when AI systems produces unfair, inaccurate, or discriminatory results because of data it was trained on or the way it was designed.

- An AI model is considered biased when it unfairly favours or disadvantages certain people or groups.

Bias usually occurs because:
→ Traing data is unbalanced.
→ Historical data contains discrimination.
→ Important groups are underpresented.
→ Human assumptions influence the system.

A well known example: ~~of biased is a~~ AI ~~hiring~~
* Biased AI hiring systems.
→ Some hiring systems/AI tools were trained using past employee data. If the company historically hired more men than women, the AI would have learned to favour male candidates and downgrade applications from women.

The Consequences
→ Unfair hiring decisions
→ Discrimination against certain groups.
→ loss of trust in AI systems
→ Harm to people's careers or opportunities

(33)

ROLE	MAIN RESPONSIBILITY	KEY SKILLS	COMMON TOOLS
DATA ANALYST	Analyses data to find trends and create create reports for business decisions	Data analysis, SQL, reporting and visualisation	Microsoft excel, Microsoft Power BI, Tableau
DATA SCIENTIST	Builds predictive models and uses advanced analytics to solve complex problems	Machine learning, statistics, python data modelling	Python, Jupyter notebook, Scikit-learn
DATA ENGINEER	Designs and maintains systems that collect, store and process data	Databases, ETL, cloud, computing, big data	Apache Spark, My SQL, Amazon Web Services
AI ENGINEER	~~Designs~~ Develops and deploys AI and machine learning systems	Deep learning, AI models, Programming and neural networks	TensorFlow, PyTorch, ChatGPT APIS

① Data Analysts
- focus on reporting and understanding data.

② Data Scientists
- Build predictive and machine learning models.

③ Data engineers
- create and manage data infastructure

④ AI Engineers
- develop intelligent AI systems and applications

(34) Business Intelligence (BI)
- is the process of collecting, analysing and presenting data to help Businesses make informed decisions

- BI uses
* → Reports
* → Dashboards
* → Charts
* → Data visualisations

to turn raw data into useful business insights. Businesses use (BI) to track performance, identify trends, improve decision making, increase profits and solve business problems.

Popular BI TOOLS

① Microsoft Powe BI
- a data visualsation and reporting tool developed by microsoft

② Tableau
- A powerful analytics and visualisation platform.

Examples

* A retail company can use BI tools to identify which products sell the most

* A bank can monitor customer spending patterns

SELF REFLECTION

① The ~~most~~ concept I found most interesting was Artificial Intelligence and Machine learning. I was surprised by how AI systems can learn patterns from data and improve overtime without being directly programmed for every task. I also found it interesting how AI is already part of everyday life through apps like ChatGPT, recommendation systems, voice assistance and self-driving car technology. Learning how AI can analyse data and make decisions showed me how powerful technology has become and much it will shape the future.

② The concept I found most difficult to understand was deep learning and Neural Networks because it was challenging to understand how machines process information similarly to the human brain. To understand it better, I used online explanations, videos, examples and simplified notes. I also used AI tools ~~AI~~ like ChatGPT to break the concepts into simpler explanations and real-world examples, which made the topic easier to understand.

(3) I believe data and AI will have major impact on both my future career and everyday life. In a future work enviroment, AI tools can help auto-mate tasks, analyse data faster, and improve decision-making. For example, businesses use Business Intelligence tools and AI-systems to track performance, predict trends and improve customer-service.

In everyday life AI already affects how people use smart-phones, social media, navigation apps and online-shopping. For example, Google-Maps uses AI to predict traffic and suggest faster routes, while streaming platforms reccommend movies based on viewing history. I believe learning about data and AI is important because these technologies will continue growing and becoming part of many careers and daily activities.